

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Privacy Enhancing Techniques

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C, C#, SQL

Data Science: Python, Pandas, NumPy, Exploratory Data Analysis, Feature Engineering, Predictive Modeling, Statistical Analysis, Anomaly Detection, Data Visualization, Model Evaluation

Databricks: Databricks, Apache Spark, PySpark, Spark SQL, Delta Lake, Databricks SQL, Lakehouse Architecture

Data Engineering: ETL, ELT, Data Pipelines, Data Ingestion, Data Extraction, Data Transformation, Data Processing, Data Validation, Data Cleaning, Data Quality, Batch Processing, Real-Time Processing, Workflow Automation

Streaming: Apache Kafka, Apache Flink, Flink SQL, Stream Processing, Event-Time Processing, Windowing, Watermarking, Real-Time Analytics

Databases: SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, ClickHouse, NoSQL, OLAP

Cloud: AWS, Glue, EMR, Lambda, Amazon S3, Amazon EC2, Amazon RDS, Docker, Linux

Tools: Git, GitHub, Grafana, Streamlit, Jira, Agile, Scrum

PROFESSIONAL EXPERIENCE

Data Engineering Intern, InsOps Inc., Chicago, Illinois

Jul 2024 – Present

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, Apache Airflow for orchestration and cloud storage in S3 for scalable data ingestion, transformation, validation, and analytics.
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics.
- Implemented metadata driven data processing for automated schema discovery, transformation, and data validation across heterogeneous sources.
- Applied data cleaning, transformation, feature preparation, and quality validation to create reliable, analytics-ready datasets.
- Worked with distributed data processing and lakehouse concepts, including scalable ETL/ELT, cloud data storage, SQL transformations, and pipeline orchestration.

Teaching Assistant, Department of Computer Science and Engineering, University of Arkansas

Sep 2024 – May 2026

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals
- Assisted students with control flow, functions, arrays, programming logic, debugging, and problem-solving techniques
- Supported Database Management Systems instruction for approximately 180 students, assisting with database concepts, assignments, assessments, and technical questions

Project Engineer, Soliton Technologies, India

Jul 2022 – Jun 2024

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Implemented reusable software components and automated workflows within an object-oriented application architecture, improving development and testing efficiency
- Developed customized software solutions for Medtronic as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements
- Integrated automated testing and validation workflows to improve test coverage, reduce manual execution, and support repeatable software verification
- Collaborated with cross-functional engineering teams to debug system-level issues, troubleshoot application behavior, optimize workflows, and deliver production-ready software

PROJECTS

Real-Time Medical IoT Analytics Platform

Apache Kafka, Apache Flink, SQL, Python, ClickHouse, Grafana

- Developed an end-to-end, sub-second latency streaming platform using Apache Flink SQL and Apache Kafka to ingest, process, and score high-frequency medical IoT device data
- Implemented temporal Flink OVER windows partitioned by device ID to dynamically compute rolling Z-scores over 5-second sliding intervals, applying custom watermark delay strategies to cleanly handle out-of-order data streams
- Designed high-performance ClickHouse schemas using the MergeTree engine sorted by composite keys to physically align disk layouts with clinical querying patterns, while isolating ingestion via an optimized Python multi-topic batch consumer
- Integrated a Grafana visualization layer directly on top of the ClickHouse OLAP tier, delivering real-time clinical dashboards that monitor raw ventilator telemetry and instantly surface streaming Z-score anomalies